

Software Engineering

How To Create Software Products Everyone Loves!

Module 12

UI/UX Fundamentals: Design Process & Developer Handoff

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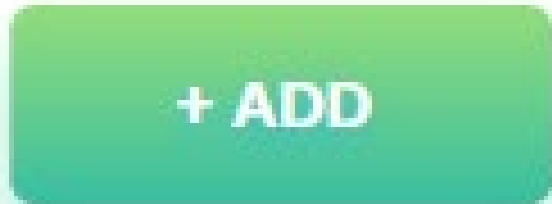
- 1 Design Process
- 2 Developer Handoff
- 3 Prototyping

Keep Your Design Files Tidy

Even when doing design exercises, you need to make sure your design files are organized and tidy.

Components

No matter where in the world you are, name your components and groups in English.



CTA - BUTTON - SMALL - GREEN GRADIENT

CTA / BUTTON / SMALL / GREEN GRADIENT

TOP LEVEL SECTIONS

When you've named all the individual objects (like icons, buttons, windows), it's time to combine them all into groups.

Dashboard

-  Header
-  Navigation
 -  Menu
 -  User account
-  Content
 -  Graph
 -  Product card
 -  Interactions
-  Footer

TEXT STYLES

After designing the major screens of your project, it's time to start creating the text styles.

What we do?

HEADING 1

Projects from 2017-2019

BODY COPY

[Read more](#)

LINK

[Read more](#)

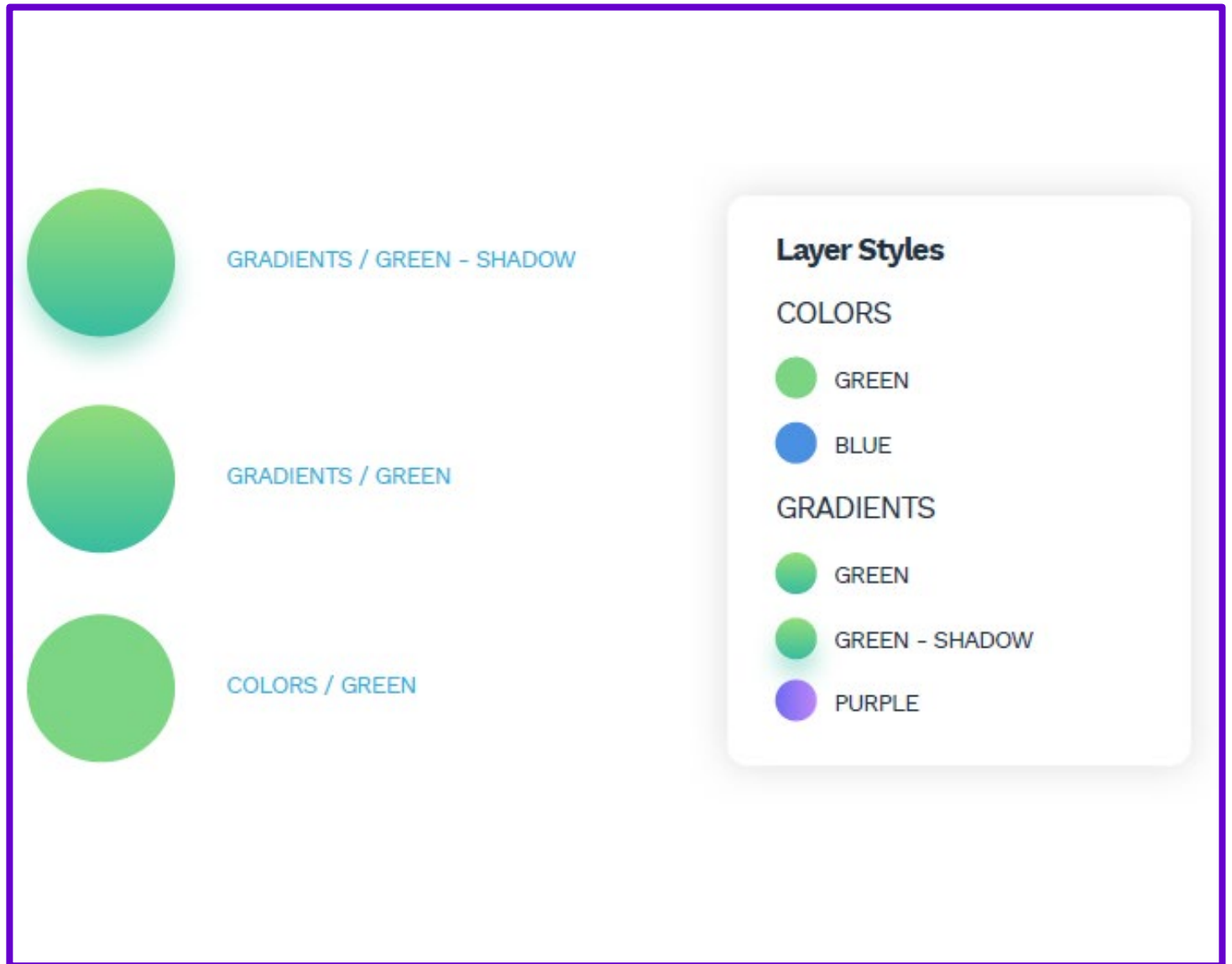
LINK HOVER

HOW TO NAME YOUR TEXT STYLES?

The best naming convention comes from HTML and CSS. Name your headers in the way they work in code (H1, H2, H3, H4). The same goes for your paragraphs (P), and your links.

LAYER STYLES

Layer styles save the properties of the object under a common name. They include the fill (color or gradient), stroke (or no-stroke), shadows, opacity, and blurs.



STYLEGUIDES AND DEFINITIONS

You can showcase your styles with a bit of CSS code on a separate artboard, to have a very early attempt at a design system.

Typography

Heading 1

```
font-family: WorkSans-ExtraBold;  
font-size: 20px;  
color: #1E3040;  
letter-spacing: -1px;  
line-height: 15px;
```

Body Copy

```
font-family: WorkSans-Regular;  
font-size: 11px;  
color: #1E3040;  
letter-spacing: -0.55px;  
line-height: 15px;
```

STYLEGUIDES AND DEFINITIONS

You can showcase your styles with a bit of CSS code on a separate artboard, to have a very early attempt at a design system.

Colors and Gradients



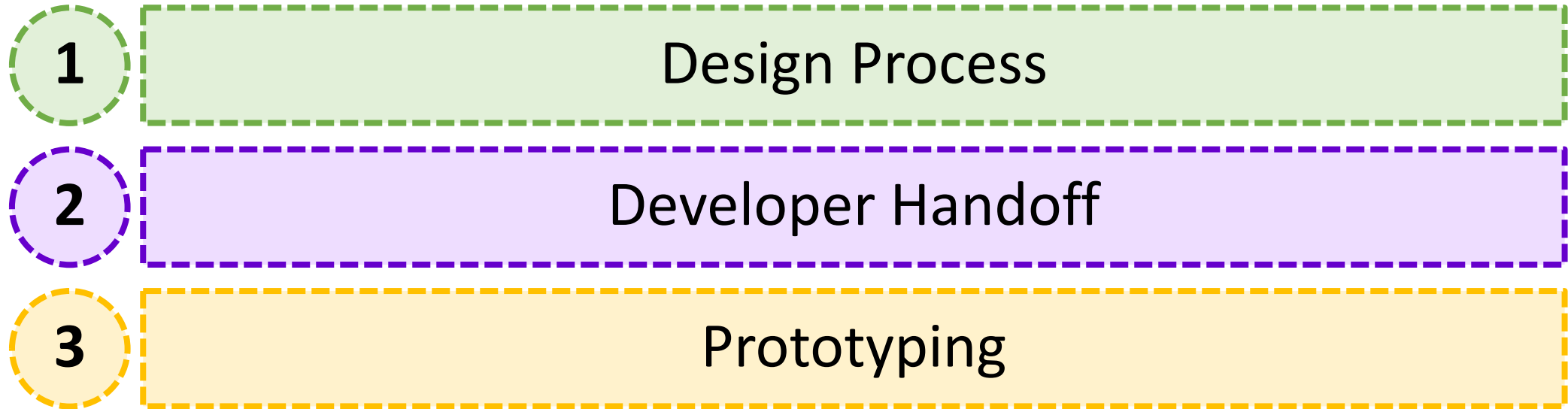
GRADIENTS / GREEN - SHADOW

```
background-image: linear-gradient(0deg,  
#13B0B0 0%, #F4FF51 100%);  
box-shadow: 0 5px 7px 0  
rgba(63,192,158,0.40);
```



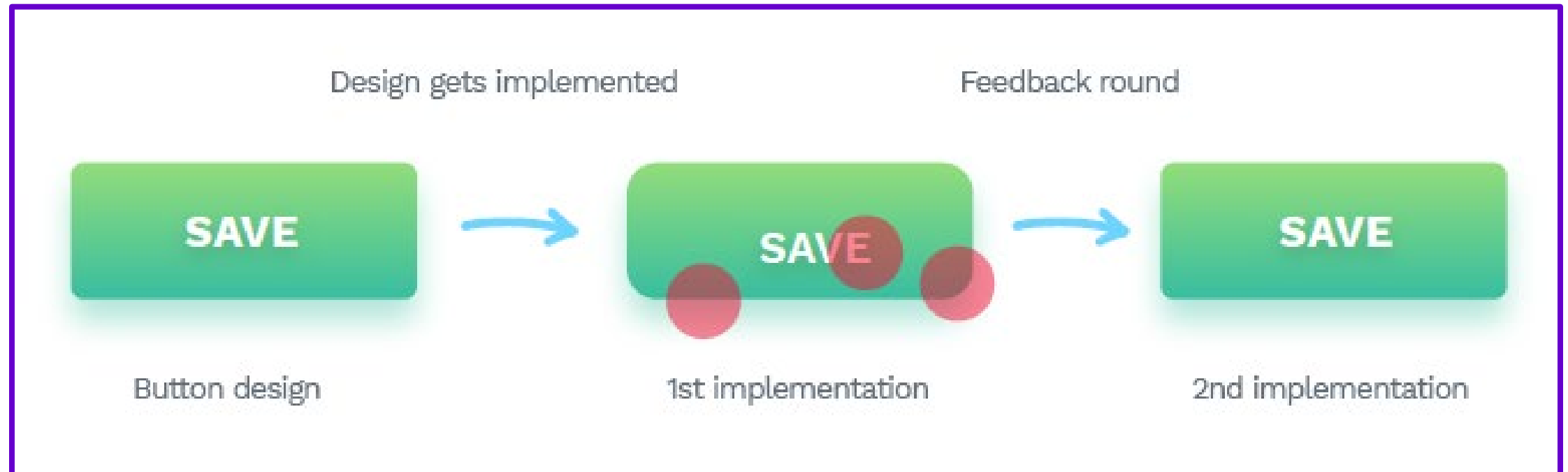
COLORS / GREEN

```
background: #7CD584;
```



How to Work with Developers

Designers should play a vital role in the QA (Quality Assurance) process by evaluating the precision of the implementation, testing, and tweaking the micro-interactions.



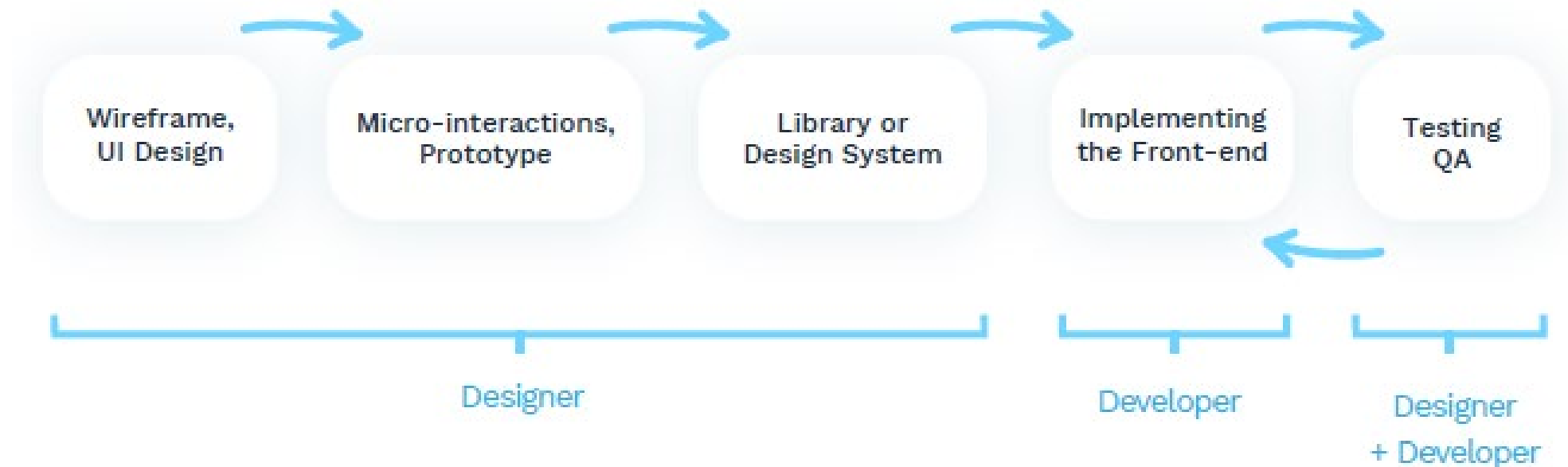
Designers Should Collaborate with Developers

When we test a real, coded screen of our app or website **on the actual device**, some tweaks may be necessary.

That's why collaboration between designers and developers is essential **even after the design process is complete.**

Development And The Process

The development consists of two main parts (using a vast oversimplification for the sake of clarity):



LIBRARIES AND DESIGN SYSTEMS

The design library is a separate file, with all of the components from the main project.



CSS of this button

```
background-image: linear-  
gradient(0deg, #13B0B0  
0%, #F4FF51 100%);  
box-shadow: 0 5px 7px 0  
rgba(63,192,158,0.40);  
border-radius: 3px;
```

```
padding: 0 24px;  
margin: 8px;
```

```
font-family: Work-Sans;  
font-size: 10px;  
color: #FFFFFF;  
letter-spacing: 0.15px;  
text-align: center;  
text-shadow: 0 2px 4px  
rgba(0,0,0,0.12);  
line-height: 44px;
```

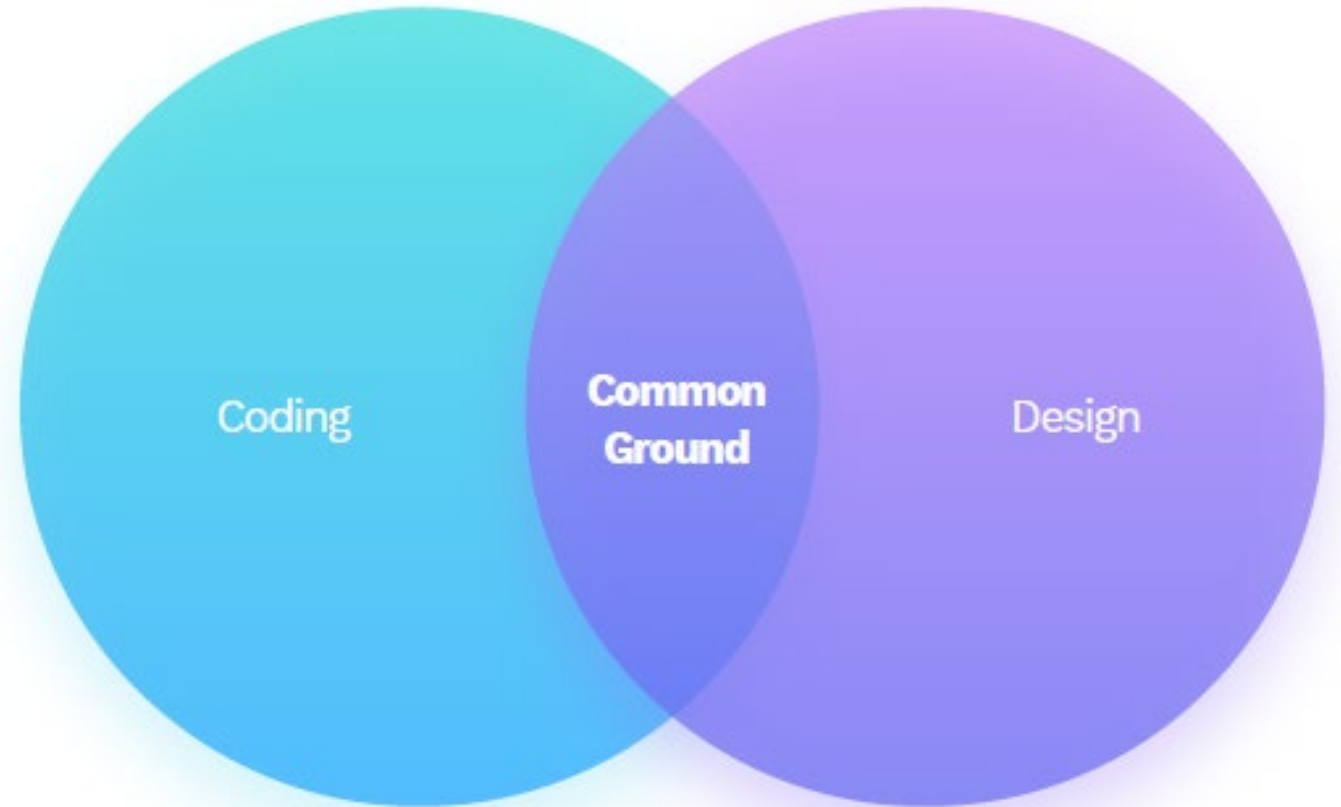
THE TOOLS

The next step is transferring both our library and all the individual screens into a **developer-friendly** tool.



Common Ground

A developer who understands design and a designer who knows a little development is the best combination for creating impressive user interfaces.



- 1 Design Process
- 2 Developer Handoff
- 3 Prototyping

Prototyping Basics

Interactive prototypes are a **simulation of how the finished product will work**

There are two main types of prototypes:

- **code-less** (inVision, Sketch, Figma)
- **fully coded** (UXPin, Framer, HTML/CSS prototypes, Axure)

Prototyping Basics

The **code-less** prototypes most often rely on image-maps, an old HTML technique of **creating invisible links on top of static images**. Those links lead to other images, **giving an illusion of a working interface**.

Prototyping Basics

The **fully coded** prototypes are a lot more advanced, as they have most of the objects **built-in code with full interactivity**. In some cases, you can also **apply additional logic** to them or even **use external data sources**.

Prototyping Basics

Prototypes also used to be divided into **low-fidelity (wireframes)** and **high-fidelity (visual UI designs)**.

lo-fi prototypes **not good for user testing** purposes. Usually Product Managers and UX designer use lo-fi in the process of feature engineering.

WHY SHOULD YOU PROTOTYPE?

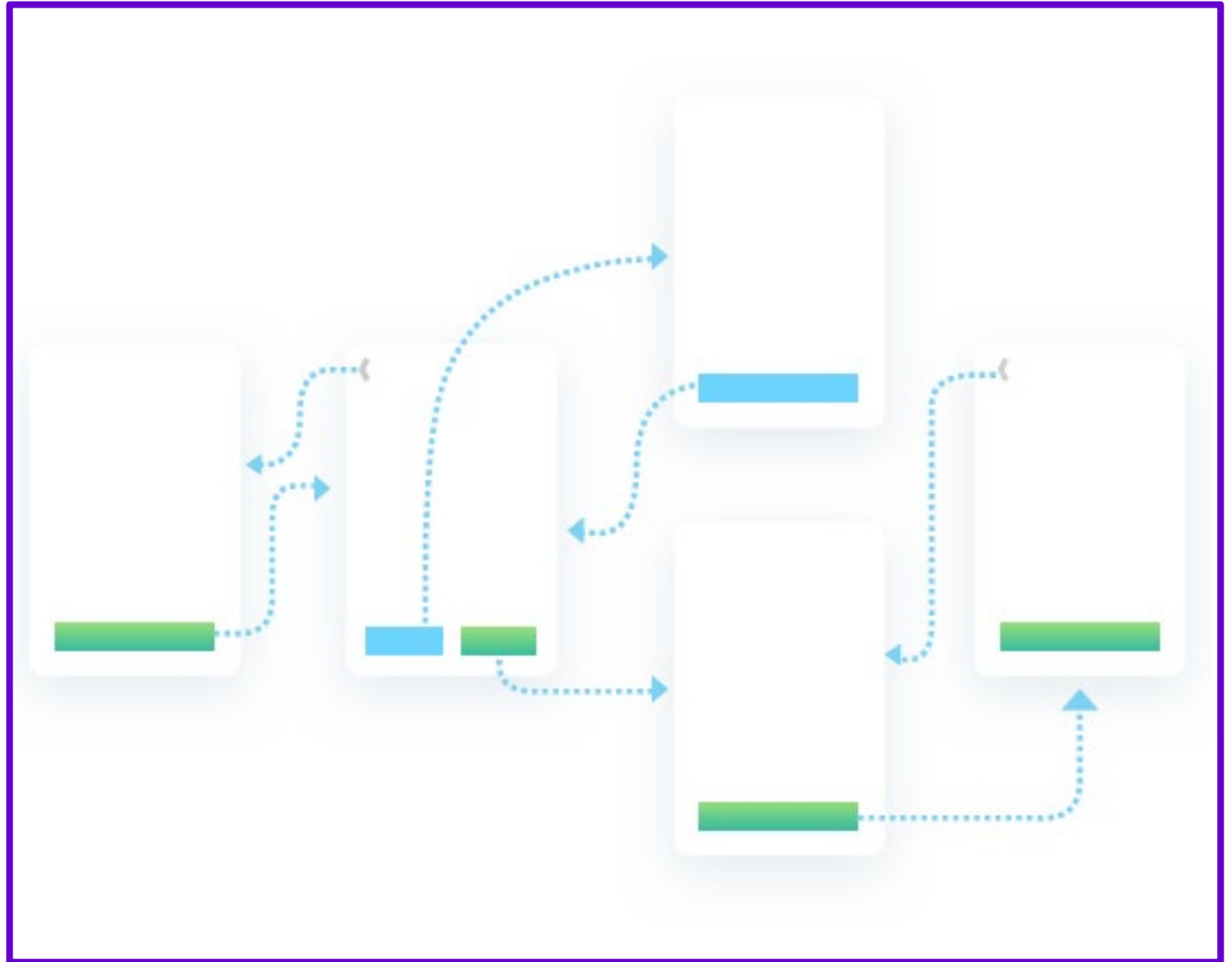
When designing new, exciting features for a product,
the "how" of the entire process is mostly in your head.

Sure, you can document of "how", but that can lead to
the famous "*game of telephone*" problem:
not everyone understands you the same way

It's a lot easier to **show someone how a product works**
than to explain it in writing.

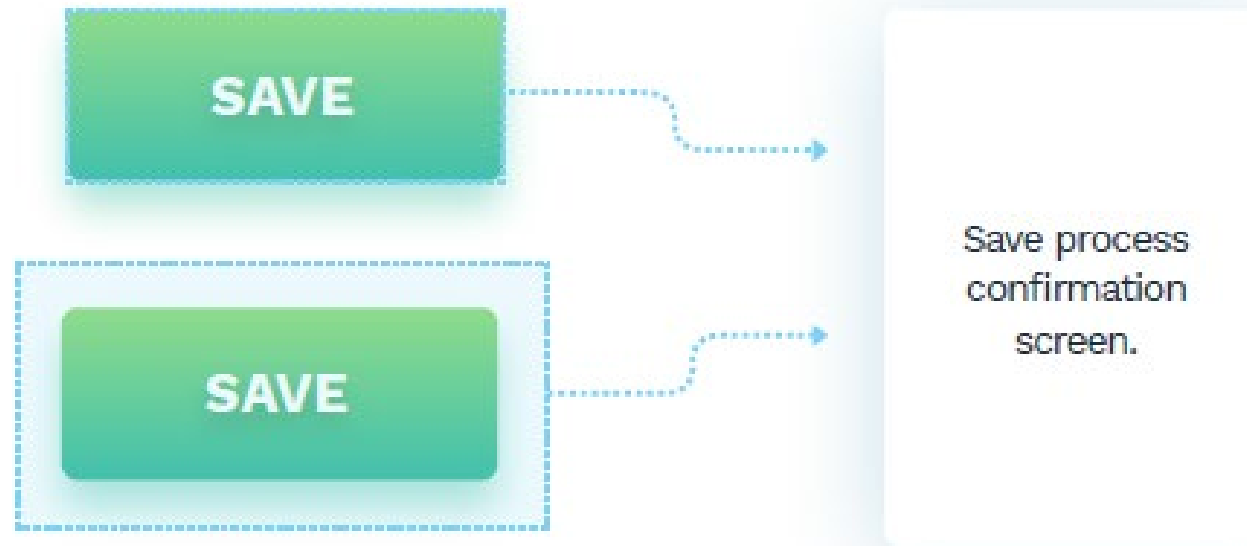
FLOW DIAGRAMS

Most code-less design tools display prototypes as flow-diagrams.



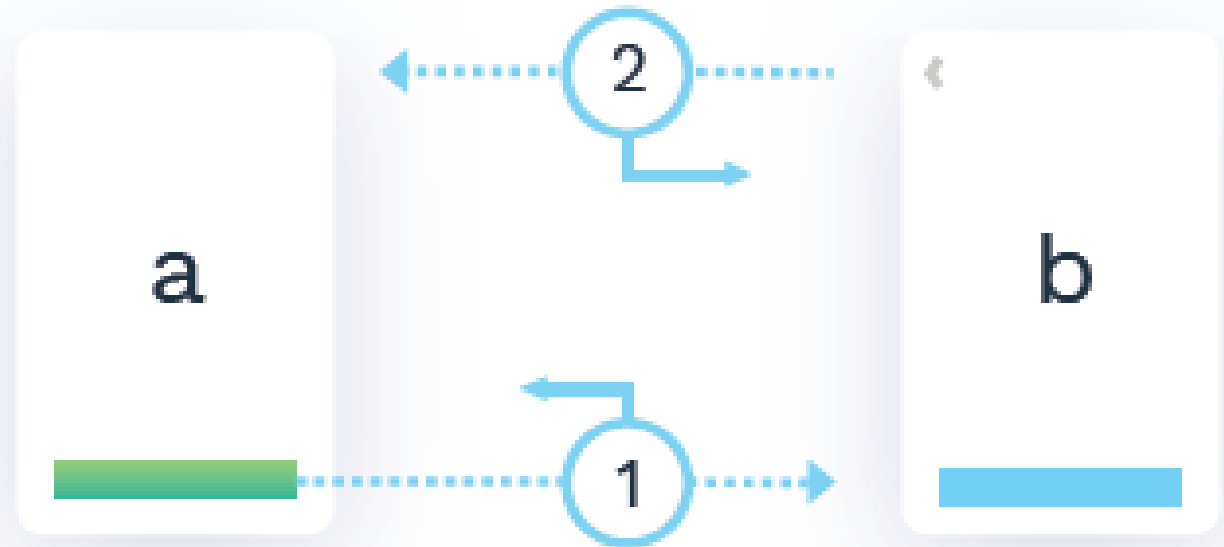
CODE-LESS PROTOTYPES

Code-less prototypes are very fast to build and a great way to show the product to stakeholders while the design process still goes on.



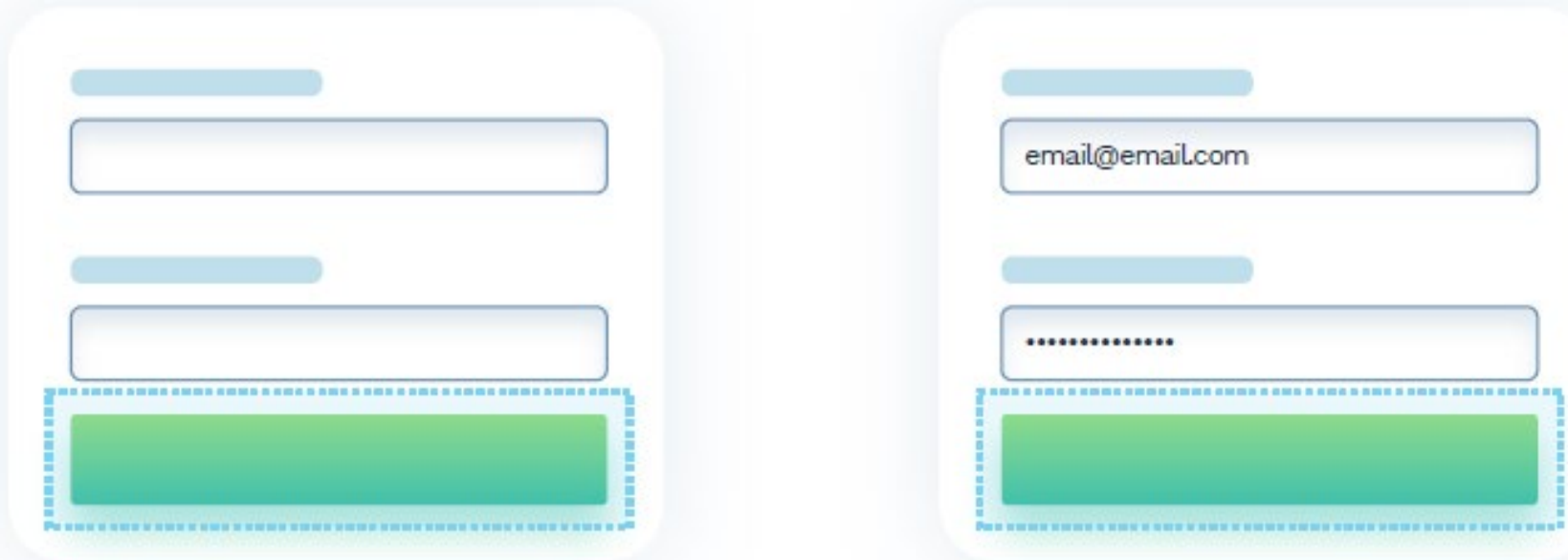
TRANSITION TYPES

The general rule is for the transition to show where each screen exists in space.



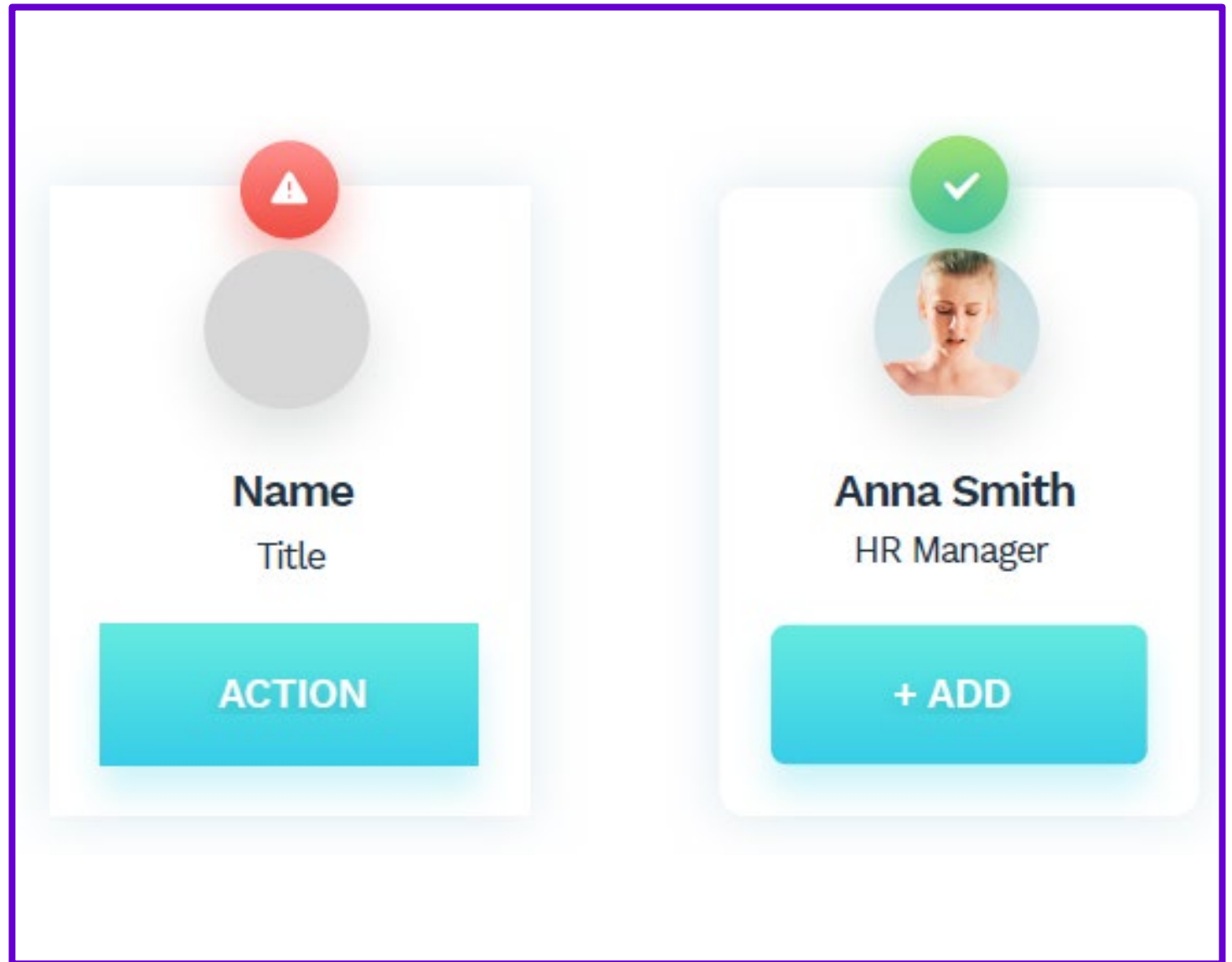
Coded Prototypes

The aim is to be able to take those already pre-built high fidelity components and use them in the final product.



REAL DATA

When creating prototypes, avoid Lorem Ipsum and dummy images.



RESEARCH

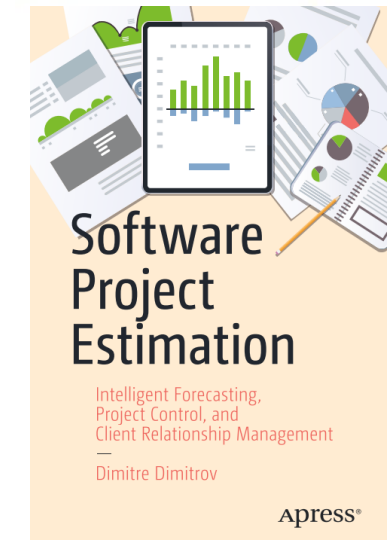
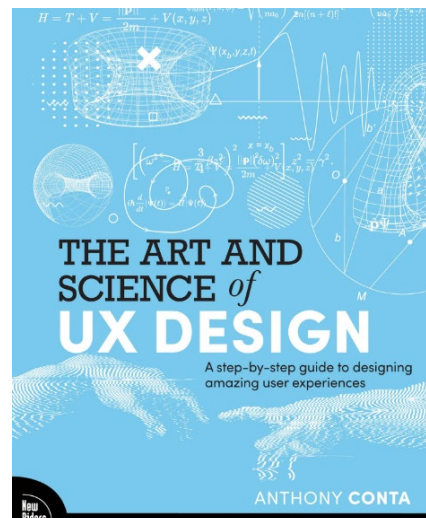
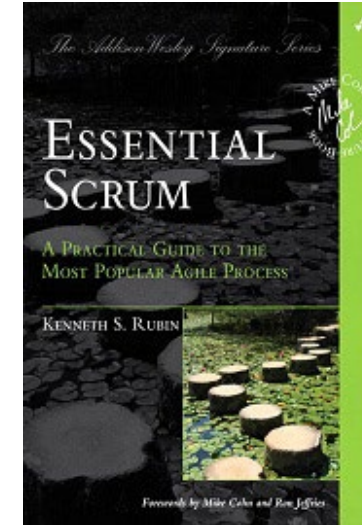
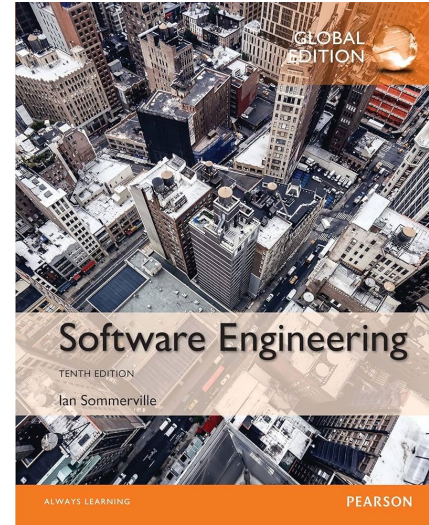
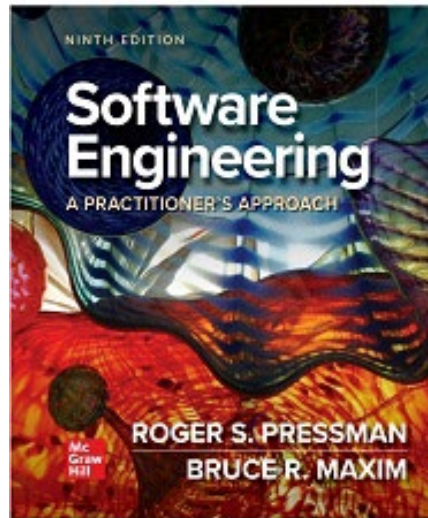
The whole concept behind prototyping is to be quick and **agile**.

**Interactive
Prototype**



**Testing
and research**

Course References



Course References

- ***Designing User Interfaces***, Michal Malewicz & Diana Malewice, 2020
- ***UI Design Styles: Trends and Design Patterns***, Michal Malewicz & Diana Malewice, 2020
- ***What UX Is Really About :Introducing a Mindset for Great Experiences***, Celia Hodent, CRC Press, 2022
- ***Lean UX: Designing Great Products with Agile Teams 3rd Edition***, Jeff Gothelf & Josh Seiden, O'Reilly, 2021
- ***Laws of UX: Using Psychology to Design Better Products & Services***, Jon Yablonski, O'Reilly, 2020
- ***Designing and Prototyping Interfaces with Figma***, Fabio Staiano, Packet Publishing, 2022

Course References

- [1] Green, M. D. (2016). Scrum: Novice to Ninja: Methods for Agile, Powerful Development, SitePoint.
- [2] Ockerman, S. and S. Reindl (2019). Mastering professional scrum: A practitioner's guide to overcoming challenges and maximizing the benefits of agility, Addison-Wesley Professional.
- [3] Martin, R. C. (2019). Clean Agile, Pearson Education.
- [4] Hall, G. M. (2017). Adaptive Code: Agile coding with design patterns and SOLID principles, Microsoft
- [۵] سامانی‌پور، علی. (۲۰۱۸). آموزش اسکرام برای مدیریت چابک فرایند توسعه اپلیکیشن های وب و موبایل،
فرادرس